

**Table 4:** Hiring, Separations, and Promotions

|                                       | Hiring               | Separation           | Promotion (rate)   |
|---------------------------------------|----------------------|----------------------|--------------------|
| <b>Treat <math>\times</math> Post</b> | −4.006***<br>(0.221) | −1.089***<br>(0.163) | 0.033**<br>(0.014) |
| Pre-period mean                       | 5.059                | 4.154                | 0.822              |
| Observations                          | 7,941,179            | 7,941,179            | 7,690,581          |
| Clusters (firms)                      |                      | 280,866              |                    |

*Notes:* This table reports the estimated  $\beta$  from Equation 15. Estimation begins in 2018 to exclude early-sample trends. Standard errors clustered by firm in parentheses.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## 6 Conclusion

This paper provides early, large-scale evidence that the diffusion of GenAI since 2023 is associated with *seniority-biased* employment effects within firms. Using résumé and job-posting data linked to roughly 281,000 U.S. firms, together with a firm-level adoption measure based on “GenAI integrator” vacancies, we document a decline in junior employment relative to senior employment at adopting firms compared with non-adopting firms. This decline is concentrated in high-exposure occupations: within adopting firms, junior employment in high-exposure occupations contracts relative to low-exposure occupations after 2022, while the corresponding senior series continues to rise. Task-level evidence from job postings points to the same mechanism: GenAI-exposed tasks become less likely to appear in junior task bundles at adopting firms. Finally, decomposing employment flows shows that the decline is driven primarily by slower hiring rather than increased separations.

These findings should be interpreted with caution. Adopting firms differ systematically in size, workforce composition, and industry. Although our research designs absorb many observable and unobservable differences across firms, sectors, and time, unobserved confounding factors may remain. We therefore examine four prominent alternative explanations: monetary tightening, post-pandemic hiring dynamics, the expansion of remote work, and labor-supply shocks. These tests support our interpretation, but they cannot rule out all alternative explanations. In addition, our adoption measure captures deliberate organizational integration of GenAI but may miss informal or “silent” adop-

tion within firms. Finally, because GenAI is still in the early stages of diffusion, longer-run adjustments in training, task allocation, wages, and internal career ladders may either attenuate or amplify the effects documented here.

Even with these caveats, the results suggest that GenAI adoption may be shifting labor demand away from entry-level tasks and narrowing the lower rungs of internal career ladders. Because early-career jobs are central to skill formation and lifetime earnings growth, such changes may have consequences beyond short-run employment. Whether these patterns persist, and whether firms respond by redesigning training, tasks, or career progression, remains an important question for future research.

## References

- Acemoglu, D. and Autor, D. (2011). Skills, Tasks and Technologies: Implications for Employment and Earnings. In *Handbook of Labor Economics*, volume 4, pages 1043–1171. Elsevier.
- Acemoglu, D., Autor, D., Hazell, J., and Restrepo, P. (2022). Artificial Intelligence and Jobs: Evidence from Online Vacancies. *Journal of Labor Economics*, 40(S1):S293–S340.
- Acemoglu, D. and Restrepo, P. (2018). The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment. *American Economic Review*, 108(6):1488–1542.
- Acemoglu, D. and Restrepo, P. (2019). Automation and New Tasks: How Technology Displaces and Reinstates Labor. *Journal of Economic Perspectives*, 33(2):3–30.
- Acemoglu, D. and Restrepo, P. (2020). Robots and Jobs: Evidence from U.S. Labor Markets. *Journal of Political Economy*, 128(6):2188–2244.
- Acemoglu, D. and Restrepo, P. (2022). Tasks, Automation, and the Rise in US Wage Inequality. *Econometrica*, 90(5):1973–2016.
- Althoff, L. and Reichardt, H. (2026). Task-Specific Technical Change and Comparative Advantage.
- Appel, R., Massenkoff, M., McCrory, P., McCain, M., Heller, R., Neylon, T., and Tamkin, A. (2026). The Anthropic Economic Index report: Economic Primitives.
- Autor, D. (2024). Applying AI to Rebuild Middle Class Jobs. Technical report, National Bureau of Economic Research.
- Autor, D., Levy, F., and Murnane, R. J. (2003). The Skill Content of Recent Technological Change: An Empirical Exploration. *The Quarterly Journal of Economics*, 118(4):1279–1333.
- Autor, D. H. and Dorn, D. (2013). The Growth of Low-Skill Service Jobs and the Polarization of the U.S. Labor Market. *American Economic Review*, 103(5):1553–1597.
- Azar, J., Gine, M., and Sanz-Espín, J. (2025). AI Is Already Eroding Wages: Quasi-Experimental Evidence From Occupational Exposure. Technical report.
- Babina, T., Fedyk, A., He, A., and Hodson, J. (2024). Artificial Intelligence, Firm Growth, and Product Innovation. *Journal of Financial Economics*, 151:103745.